

Final Year Project Document

Multi-Agent Market Research Platform

Chapter 2 — Background and Literature Review

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Chapter 2

Background and Literature Review

A structured literature review establishes the foundation for modern software engineering research and development. Comparative analysis identifies practical shortcomings across manual survey platforms and clarifies why an automated architecture is necessary. Software benchmarking positions ongoing development within the wider computing discipline, spotlighting unresolved operational bottlenecks and setting concrete technical boundaries for engineering choices. This chapter systematically evaluates relevant academic publications alongside commercial mobile survey systems to substantiate the design of the proposed platform [4].

A clear roadmap will be followed throughout this chapter. In Section 2.1, capabilities and limitations of existing survey platforms will be evaluated. Practical field usage of a future platform will be examined in Section 2.2, from which the formal problem statement will be formulated in Section 2.3. The proposed market research platform will be described in Section 2.4, followed by a justification of the selected software process model in Section 2.5.

2.1 Existing System

Software engineers conduct rigorous comparative evaluations of existing commercial platforms to ground new system development. Systematic product benchmarking identifies standard operational features across deployed survey tools while exposing unaddressed technical shortcomings in field workflows. Software benchmarking positions ongoing development within the wider computing discipline, spotlighting unresolved operational bottlenecks and setting concrete technical boundaries for engineering choices. Investigating production software architectures prevents development teams from repeating known design flaws and reinventing established engineering mechanisms. Understanding these functional boundaries establishes an empirical justification for building specialized software architectures. This section systematically evaluates ten widely used data collection platforms, analyzing offline SQLite storage, spatial billboard registries, and multi-agent competitive workflows [4], [5].

Ten systems are reviewed in this section: Qualtrics, SurveyMonkey, ODK, KoboToolbox, Fulcrum, SurveyCTO, CommCare, FastField, GeoPoll, and QuickTapSurvey. Selected as established market leaders and open-source standards, these applications enable survey authoring and offline field collection. Despite their individual strengths, they consistently lack multi-agent reasoning, automated billboard evidence verification, and

geofenced visit controls required for advertising research. A detailed comparison table summarizes these systems at the end of the section, followed by an evaluation of common patterns and unaddressed gaps that transitions to the next section.

Qualtrics: Qualtrics provides an enterprise-grade experience management and surveying suite that organizations deploy for customer feedback and academic research. The platform features an advanced web-based questionnaire builder equipped with conditional branch logic, quota tracking, and automated response reporting. Organizations distribute digital surveys through web links, scheduled emails, and native mobile client applications. The software stores survey submissions in cloud database clusters and delivers analytics through configurable operational dashboards. While Qualtrics excels at complex psychometric survey designs and longitudinal consumer monitoring, its architecture exhibits clear operational constraints in field research environments. The mobile client does not maintain an offline-first SQLite synchronization queue with incremental retry handling for remote areas. The platform lacks automated artificial intelligence workflows for billboard image quality assessment, physical site verification, and autonomous agent orchestration [1].

The main features of the Application are listed below:

- Advanced web-based questionnaire builder with complex branching logic
- Automated quota management and digital distribution pipelines
- Enterprise role-based access control and respondent privacy governance
- Real-time analytical dashboards with statistical cross-tabulations

The limitations of the application are listed below:

- No offline-first SQLite synchronization queues with retry policies
- Absence of geofenced attendance verification for field workers
- No computer vision models for billboard visibility or image quality grading
- Lacks autonomous multi-agent orchestration for secondary market research

SurveyMonkey: SurveyMonkey operates as a widely adopted cloud surveying platform that enables organizations to design, distribute, and analyze standardized web questionnaires. The system provides an intuitive browser interface alongside mobile collection utilities, allowing users to configure response validation rules, logic branching, and basic demographic quotas. Submissions aggregate within centralized cloud storage, where built-in analytics modules generate summary charts, sentiment scores, and filtered cross-tabulations. SurveyMonkey serves commercial market research teams and corporate marketing departments seeking rapid customer feedback. The system presents distinct technical limitations for physical field investigations and out-of-home advertising verification.

The platform lacks hierarchical location registries linked to specific field personnel and does not support geofenced attendance validation. SurveyMonkey offers no native computer vision tools for evaluating billboard media or multi-agent pipelines for secondary market intelligence [2].

The main features of the Application are listed below:

- Intuitive browser-based questionnaire design interface
- Pre-built survey templates and standard demographic question banks
- Automated response summarization and basic sentiment scoring
- Multi-channel survey distribution via web, email, and social links

The limitations of the application are listed below:

- Lacks hierarchical location registries for physical billboards
- No geofenced attendance logging or street-level visit verification
- No local database synchronization queues for unstable field connections
- No multi-agent research planning or automated cross-source evidence checks

ODK: Open Data Kit provides an open-source data collection platform for field research teams operating across regions with minimal network infrastructure. The platform pairs a secure central web server repository with a native Android client application. Field workers download digital questionnaires onto mobile devices to gather respondent feedback and survey answers without active internet connectivity. The mobile client records precise geospatial coordinates, captures site photographs, and stores field submissions locally until wireless connectivity permits server synchronization. Researchers and humanitarian teams deploy this platform for remote enumeration, demographic studies, and continuous environmental monitoring. The software exhibits notable architectural limitations for commercial billboard advertising audits and spatial market research. The platform lacks built-in artificial intelligence agents, cross-source evidence checks, and automatic market intelligence report generation [3].

The main features of the Application are listed below:

- Standardized XForms support for complex offline mobile questionnaires
- Android field data collection with GPS logging and photo capture
- Open-source backend architecture with self-hosting capabilities
- Asynchronous batch synchronization over cellular or Wi-Fi networks

The limitations of the application are listed below:

- No hierarchical campaign management or billboard registry
- No geofenced attendance enforcement tied to planned site visits
- No automated image quality or billboard visibility assessment
- No multi-agent generative reasoning for competitor market analysis

KoBoToolbox: KoBoToolbox serves as an open-source data collection platform built for humanitarian operations and academic research teams. The system combines an online questionnaire creator with native mobile collection tools to facilitate systematic field investigations. Field personnel deploy the mobile client across remote survey zones to record questionnaires, register GPS points, and store site photographs without active internet connectivity. The platform aggregates field submissions onto a central repository where supervisors inspect response charts, examine spatial distribution maps, and export raw data tables. Humanitarian agencies rely on this platform for rapid population census runs, crisis monitoring, and community health evaluations. The software shows noticeable architectural limitations when organizations attempt commercial advertising research. The platform lacks custom registries for physical billboard inventory, geofence boundary attendance tracking for audit staff, and automatic artificial intelligence workflows for assessing billboard visibility [6].

The main features of the Application are listed below:

- Intuitive form builder supporting standard ODK and XForms protocols
- Offline field survey data collection on mobile devices
- Central web dashboard with real-time response aggregation and map views
- Role-based user permissions and granular workspace sharing settings

The limitations of the application are listed below:

- No dedicated registry for billboard structures or outdoor marketing assets
- Absence of geofenced attendance verification for assigned field surveyors
- Lacks automated computer vision routines for billboard obstruction scoring
- No autonomous multi-agent orchestration for competitive market intelligence

Fulcrum: Fulcrum operates as a location-first mobile inspection platform built for field teams managing spatial asset inventories. The software allows organizations to craft custom survey forms that integrate spatial mapping layers and photographic evidence collection. Mobile workers utilize the application to document infrastructure condition, record geographic coordinates, and capture high-resolution imagery in offline field environments. The system maintains local data persistence and uploads pending inspection

records automatically when devices regain wireless network access. Engineering firms and utility providers deploy this platform for asset inspections, environmental compliance surveys, and facility audits. The platform presents distinct architectural constraints for commercial marketing intelligence campaigns. The software omits multi-agent reasoning graphs for competitor intelligence, natural language query interfaces for research dashboards, and automatic computer vision scoring for billboard visibility [7].

The main features of the Application are listed below:

- Location-first mobile data collection with custom inspection forms
- High-accuracy GPS mapping and spatial layer integration
- Offline field data caching with automatic background synchronization
- Geotagged photographic capture and multimedia inspection attachments

The limitations of the application are listed below:

- No multi-agent architecture for autonomous competitor intelligence gathering
- Absence of automated image visibility analysis and billboard angle scoring
- Lacks conversational natural language querying for research analytics
- High enterprise subscription costs restrict deployment in smaller teams

SurveyCTO: SurveyCTO provides a secure mobile data collection suite built for rigorous field evaluations and academic research surveys. The platform pairs an online form creator with an Android collection client capable of operating in areas without cellular connectivity. Field enumerators use the mobile client to record systematic interview responses, register location markers, and upload photographs under strict end-to-end encryption standards. The software incorporates automatic data quality monitors, real-time audio audits, and statistical anomaly checks to detect fraudulent survey submissions. Research institutions and non-governmental organizations rely on this platform to maintain high data integrity during large-scale socioeconomic investigations. The software shows clear functional limitations for commercial outdoor media research. The platform omits a central registry for billboard structures, geofence boundary attendance verification for field workers, and computer vision pipelines for advertising quality assessments [8].

The main features of the Application are listed below:

- Secure mobile data collection with end-to-end encryption
- Automated quality checks and audio audit capabilities
- Offline survey completion with automated cloud synchronization
- Advanced form logic with constraint checks and validation rules

The limitations of the application are listed below:

- No native billboard asset management or spatial registry workflows
- Absence of geofenced attendance logging for physical site visits
- Lacks deep learning vision models for outdoor advertisement evaluation
- No multi-agent pipelines for automated market trend extraction

CommCare: CommCare operates as an open-source mobile platform that organizations deploy for longitudinal case management and frontline service delivery. The system allows organizational administrators to configure complex multi-step workflows, client registration forms, and decision support algorithms through a web interface. Frontline staff utilize the Android application to track individuals over successive visits, record survey responses, and capture multimedia evidence in zero-connectivity environments. The software stores survey records within local phone storage and synchronizes updates with central servers when wireless internet becomes available. Healthcare organizations and development agencies deploy this platform to coordinate community outreach and patient care monitoring. The platform exhibits fundamental structural boundaries for outdoor marketing research. The software lacks hierarchical registries for commercial advertising assets, geofence boundary attendance logging linking to campaign sites, and automatic multi-agent intelligence routines for market analytics [9].

The main features of the Application are listed below:

- Longitudinal case management and mobile survey tracking
- Offline-first mobile client operation on Android hardware
- Standardized multimedia capture with voice and image logging
- Cloud reporting portal with export tools for statistical software

The limitations of the application are listed below:

- Tailored primarily for public health rather than commercial advertising
- No computer vision workflows for automated billboard condition auditing
- Absence of autonomous multi-agent pipelines for market research tasks
- Lacks natural language query dashboards for dynamic campaign exploration

Premise Data: Premise Data operates as a community market intelligence platform that links commercial enterprises with external global contributors. The system deploys an Android smartphone application that gives discrete micro-tasks to local smartphone

users across international cities. Crowd contributors accept incoming audit tasks, photograph retail displays, record geographic coordinates, and submit questionnaire responses to confirm physical product availability. The platform verifies incoming submissions through automatic location checking, timestamp validation, and administrative review consoles before releasing contributor compensation. Global consumer goods manufacturers rely on this network to monitor retail distribution, track product pricing, and audit supermarket shelves. The software shows notable operational constraints for continuous outdoor advertising research. Premise Data lacks offline-first SQLite synchronization queues for private field teams. The software also omits automatic computer vision routines for billboard obstruction scoring and multi-agent competitive research workflows [10].

The main features of the Application are listed below:

- Crowdsourced global network for real-time micro-task survey execution
- Geotagged image submission with timestamp and coordinate validation
- Automated reward allocation and contributor payment infrastructure
- Central web analytics interface with geospatial heatmap visualization

The limitations of the application are listed below:

- Closed proprietary platform with no support for self-hosted instances
- Lacks dedicated offline SQLite queue synchronization for private audits
- No multi-agent generative reasoning for competitor intelligence analysis
- High reliance on transient crowd workers reduces data auditing consistency

Streetbees: Streetbees provides an artificial intelligence consumer research platform that gathers real-time behavioral insights through smartphone interactions. The platform presents survey questions through a conversational chat interface that encourages users to share spontaneous thoughts, photographs, and video clips. Mobile respondents interact with the application during everyday shopping, dining, and brand consumption routines across global consumer markets. The central backend analyzes incoming multimedia through natural language processing algorithms that classify emotion, consumer sentiment, and product usage context. Multinational brand managers consult the resulting analytics dashboard to identify emerging market trends, evaluate advertising reception, and guide product packaging strategies. The software exhibits fundamental operational limitations for outdoor billboard research. Streetbees does not offer systematic registries for physical advertising locations, geofence visit verification for field inspectors, or computer vision models for billboard visibility grading [11].

The main features of the Application are listed below:

- Conversational chat-style mobile surveying for consumer research
- Multimedia submission allowing smartphone video and photograph uploads
- Natural language processing for sentiment and emotion extraction
- Real-time consumer behavior feeds across international markets

The limitations of the application are listed below:

- Focuses on consumer sentiment rather than physical billboard inspections
- Absence of geofenced attendance enforcement for planned field itineraries
- No offline-first SQLite database architecture for zero-connectivity zones
- Lacks computer vision scoring for physical advertising display metrics

GeoPoll: GeoPoll operates as a mobile surveying platform that conducts remote research across emerging international markets through telecommunication channels. The system connects with millions of mobile subscribers through text messages, interactive voice phone calls, and browser links to administer uniform questionnaires. Survey respondents answer polling inquiries on standard mobile phones without requiring fast broadband connections or proprietary smartphone applications. The platform aggregates incoming survey answers in real time, delivering demographic cross-tabulations, audience measurement statistics, and sentiment summaries through an online dashboard. Media broadcasters, international development agencies, and consumer brands rely on this platform to measure radio audiences and track public opinion. The software presents obvious functional constraints for outdoor advertising research. The platform lacks on-site inspection capabilities, offline-first SQLite database synchronization for field personnel, and computer vision models for auditing physical billboard structures [12].

The main features of the Application are listed below:

- Multi-channel mobile surveying across SMS, voice calls, and web links
- Extensive respondent database covering emerging international markets
- Fast data turnaround with automated response aggregation pipelines
- Interactive campaign analytics and media audience measurement tools

The limitations of the application are listed below:

- Remote telecom polling model lacks physical on-site audit capabilities
- No on-device computer vision models for outdoor billboard grading
- Absence of offline SQLite synchronization for field survey personnel
- Lacks autonomous multi-agent graphs for cross-verifying ground truth data

Table 2.1 Table 2.1 outlines the functional characteristics of the ten reviewed tools and the proposed platform. It reviews operational differences in local caching, geofenced attendance, automated billboard scoring, and agent-based research workflows.

Table 2.1: Comparison of Existing Systems

Features	**Applications**										
	Qualtrics [1]	SurveyMonkey [2]	ODK [3]	KoBoToolbox [6]	Fulcrum [7]	SurveyCTO [8]	CommCare [9]	Premise Data [10]	Streetbees [11]	GeoPoll [12]	Proposed System
Web Questionnaires	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Offline Mobile Collection	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓
Android Field Collection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Media Capture (Photo/Audio)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
GPS & Location Logging	✗	✗	✓	✓	✓	✓	✓	✓	✗	✓	✓
Billboard & Asset Registry	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Geofenced Attendance Verification	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Offline SQLite Sync Queues	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Mobile AI Image Quality Assessment	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
AI Agent Orchestration (LangGraph)	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Human-in-the-Loop Review Pipeline	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓

2.2 Future System Usage Analysis

Market research systems will shift toward autonomous data collection and spatial intelligence across the advertising industry. Field survey platforms will incorporate cooperative artificial intelligence agents to coordinate campaign planning and competitor monitoring. Rather than relying on static questionnaires, mobile applications will run on-device computer vision models to evaluate billboard viewing angles, physical condition, and ambient lighting in real time. Smart synchronization protocols will store survey submissions locally on mobile hardware and upload batches as wireless networks emerge. Multi-agent software pipelines will automatically cross-reference street photographs with secondary web data, uncovering market trends and flagging inconsistent responses for human review. Brand strategists will consult conversational dashboards to query survey metrics and trace claims directly to physical site evidence. Through these continuous advancements, modern platforms will transform out-of-home advertising measurement into a verifiable, tamper-resistant discipline.

2.3 Problem Statement / Limitations

Outdoor advertising research currently relies on manual field auditing routines across metropolitan media markets. Survey organizations deploy field enumerators equipped with consumer smartphone applications to inspect roadside billboard structures. Field auditors record descriptive questionnaire responses, photograph commercial advertising displays, and note geographic coordinates across daily audit routes. Central analysts manually pair detached photographs with corresponding survey records before compiling advertiser campaign reports. Commercial brand managers depend on these audits to verify physical display condition, monitor competitor spending, and confirm contractual campaign delivery. The traditional auditing process exhibits severe operational limitations for large-scale media verification. Current field practices lack offline-first SQLite synchronization queues for remote survey areas, automated computer vision scoring for billboard visibility, and coordinated multi-agent workflows for competitive market intelligence.

The proposed system operates under defined practical constraints and concrete engineering limitations throughout development. The engineering team confines physical testing to regional commercial districts within Islamabad before attempting multi-city audits. The development team manages this timeline constraint by conducting structured field trials across accessible commercial hubs. Inherent variations across mobile smartphone hardware introduce data collection hurdles, as entry-level Android devices produce uneven photographic resolution and sensor noise. To resolve this hardware limitation, the backend architecture applies contrast adjustments and normalization algorithms to standardize captured photographs before automated evaluation. Furthermore, commercial outdoor rate cards remain largely proprietary and absent from open web databases. The team mitigates this data scarcity by instructing automated web agents to extract public media listings and comparing them against synthetic valuation models. The development team consciously addresses these operational boundaries across development sprints to ensure successful milestone delivery.

2.4 Proposed Solution

The proposed software architecture establishes an integrated mobile and cloud platform to automate outdoor media verification across metropolitan commercial areas. Field enumerators inspect roadside billboard structures using a native Android smartphone application operating without cellular connectivity. The mobile application records high-resolution photographs, timestamps, and geospatial coordinates within local SQLite database tables before pushing data to central servers. Central FastAPI backend services ingest incoming survey submissions and store relational campaign records within a PostgreSQL database with PostGIS extensions. LangGraph multi-agent workflows search external

advertising directories to cross-reference campaign pricing and identify competitive billboard placements. Deep convolutional neural networks evaluate captured photographs to detect physical banner tears, viewing obstruction, and poor display angles. Central administrators configure geofenced audit corridors within Islamabad to direct field teams toward unverified commercial displays. Marketing analysts examine street-level audit trails and export verification reports through interactive web consoles using natural language search queries.

2.5 Software Process Model

A software process model provides a systematic engineering framework that directs the lifecycle of modern digital systems from inception to release. This framework organizes complex engineering tasks into predictable phases, practical activities, and clear team workflows. Development teams achieve superior coordination, dependable milestone tracking, and high system reliability through such operational discipline. System characteristics such as overall scope complexity, delivery deadlines, and requirement fluidity determine the most suitable development methodology. This section outlines and defends the software process model for the system [4].

The engineering group selects Agile Scrum because developing multi-agent software demands rapid experimentation on raw media assets. Upcoming two-week sprints will produce functional Flutter components and backend APIs, while bi-weekly demos will guide the system toward timely milestone delivery.

2.5.1 Introduction

The development team adopts the Agile Scrum framework to direct system engineering across recurring cycles. This operational methodology divides complex engineering goals into distinct two-week iterations, encouraging cross-functional teamwork and early prototype evaluations. The framework suits the system because engineering autonomous research agents, offline SQLite synchronization queues, and YOLO vision algorithms demands ongoing practical trial. Software developers work across successive sprints to refine FastAPI backend services, construct Flutter mobile interfaces, and optimize multi-agent LangGraph workflows. Each development cycle begins with sprint planning, proceeds into active programming, and concludes with functional code testing and supervisor demonstrations. This rhythmic workflow helps developers resolve software defects promptly and integrate updated survey requirements without delaying scheduled milestone delivery. Figure 2.1 illustrates the complete development workflow for the system [4].

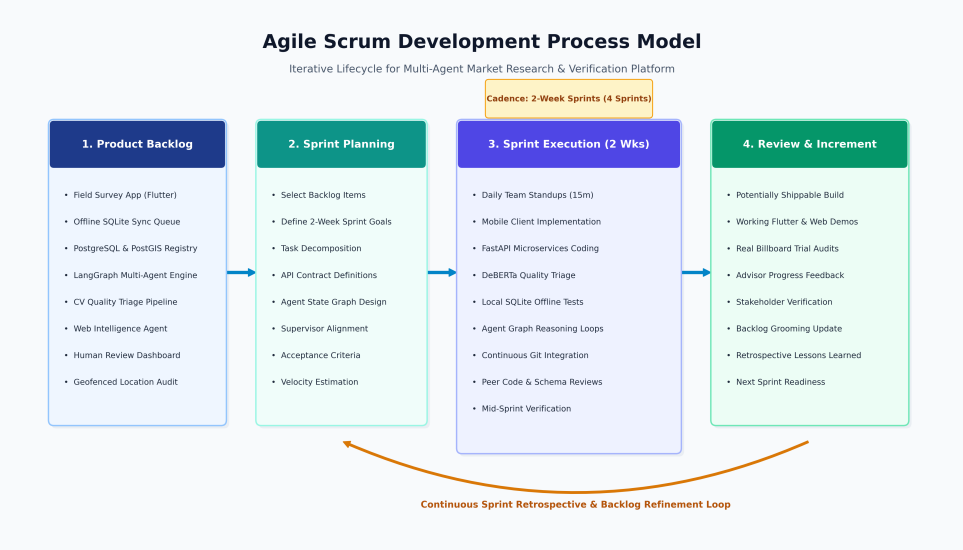


Figure 2.1: Iterative Agile Scrum development process model adopted for the system

2.5.2 Justification

The development team chooses the Agile Scrum methodology to manage engineering risks across multi-tier software modules. Building mobile survey clients, cloud spatial databases, and autonomous web agents involves tight technical interdependencies. Scrum divides these technical dependencies into manageable two-week engineering cycles, enabling developers to discover database bottlenecks early and revise agent prompts promptly. This iterative workflow allows the team to deliver stable software increments while meeting all system research requirements.

2.5.3 Steps

The development team will structure engineering milestones across four distinct Scrum iterations. Initial sprints will configure PostgreSQL schemas and local SQLite storage queues. Subsequent cycles will produce LangGraph multi-agent orchestration pipelines, computer vision quality filters, and central verification web consoles for research analysts.

Sprint 1: Core Architecture & Data Ingestion: The engineering group will build foundational database tables and authentication APIs during the opening sprint. Software developers will configure a cloud PostgreSQL instance with PostGIS geospatial modules, and the team will write baseline FastAPI endpoints to ingest incoming survey submissions from mobile clients.

Sprint 2: Mobile Client & Field Verification: Mobile engineers will construct the Flutter data collection client during the second sprint. Field staff will photograph roadside billboard structures and record coordinate markers through an Android client. The mobile application will cache inspection logs within local SQLite database files when-

ever field workers lose wireless mobile signals.

Sprint 3: Multi-Agent Intelligence & Vision Scoring: System developers will program autonomous LangGraph reasoning workflows and image evaluation algorithms during the third sprint. Multi-agent Python scripts will scrape external commercial media catalogs, while computer vision modules will inspect captured photographs for foliage obstruction, lighting glare, and angle distortion.

Sprint 4: Analytics Dashboard & System Verification: The development team will assemble central administrative consoles and conduct end-to-end integration tests during the concluding sprint. Software engineers will deploy natural language search tools for campaign analysts, and auditors will test field synchronization across several busy Islamabad commercial districts before final release.

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